Analysis of Human Tear Desiccate based on Data Analysis and Machine Learning

HackKošice 2026 – Pavol Jozef Šafárik University in Košice track

Luka Divjaković Veselin Roganović

HackKošice

April 19, 2026

Outline

1. Data Description and Preprocessing
2. Algorithms and Methods
3. Analytical Objectives
4. Expected Outcome and Results

Dataset overview

- Provided dataset: **5 folders**, one per class (chronic condition / healthy).
- Each folder contains **images** (`.bmp`) and **raw data** files.
- Dataset is **relatively small** – we searched online for additional public data without success, but studied several scientific papers on the topic.
- After exploring both modalities, we decided to use **only the `.bmp` images**.
- Pipeline: we store each image's **file path** and the **class label** (from its parent folder) in a DataFrame; the actual image is loaded on demand during training / prediction.

Patterns observed per class

- **Diabetes**: dry–leaf-like branching, many white dots, flake-like structures, sharp-angled straight lines.
- **Glaucoma**: unusual dots, droplets and spiky objects; mostly smooth curved lines, fewer sharp angles.
- **Sclerosis**: similar to diabetes but dots are more spaced; some regions are entirely white, others look flaky but are in fact clusters of dots spread everywhere.
- **Dry eye**: even more similar to diabetes; extremely regular dot groupings forming cypress-like shapes; very clean, well-defined clusters.
- **Healthy**: similar to glaucoma; more curved lines than dot clusters, occasional bright regions, cypress-like line patterns without visibly separated dots.

Data quality issues

- A portion of images are **zoomed in / at different scales** – may mislead the model.
- Presence of **duplicates and near-duplicates**.
- Some images are visibly **damaged / low quality**.
- Every image contains a **white frame and scale/metadata overlay**.

$\Rightarrow$ We removed a subset of clear outliers so the model can **generalize better**.

Preprocessing pipeline

1. **ROI crop**: remove the white frame and the scale/metadata overlay.
2. **Grayscale conversion**: color carries little signal, and our algorithms performed better on grayscale.
3. **CLAHE contrast normalization**: enhances local contrast and makes fine-grained tear-film patterns more discriminative.
4. **Data augmentation – rotations**: class patterns are not direction-specific, so rotating each image produces valid extra training samples and enlarges the dataset.
5. **Outlier filtering**: drop visibly damaged / duplicate images before training.

Evaluation protocol

- **Train / test split**: 80 / 20, stratified by class.
- **Cross-validation**: 5-fold CV on the **train** portion for model selection and hyper-parameter tuning.
- **Test set**: used only **once**, after we are happy with the model on CV, to obtain the final score.
- **Metric**: weighted F1-score (as required by the challenge).

Classical approach I – HOG + SVM / KNN

- Images clearly contain **oriented gradient patterns** (lines, dendritic shapes, dot clusters).
- **Histogram of Oriented Gradients (HOG)** captures local edge directions in cells and pools them into a feature vector.
- The HOG descriptor is then fed into classical classifiers:
  - **SVM** with RBF/linear kernel,
  - **KNN** as a simple non-parametric baseline.
- Motivation: interpretable, fast, and a natural fit for the type of texture/gradient patterns visible in the tear images.

Classical approach II – KMeans (K=3) + SVM / KNN

- Visually, most images contain roughly **3 distinct regions**:
  - background,
  - dots / main figure,
  - bright white dots.
- We use **KMeans with** K=3 on pixel intensities to segment the image into these regions.
- Statistics of the resulting clusters (area ratios, intensities, spatial descriptors) form a compact feature vector.
- This vector is classified by **SVM** or **KNN**.

Deep learning – Transfer learning with pretrained CNNs

- The dataset is too small to train a CNN from scratch.
- We use **ImageNet-pretrained** backbones, replace the classification head, and **unfreeze the last few blocks** for fine-tuning.
- Two-stage training: **head only**, then **partial fine-tuning**.
- Models tried: `ResNet18`, `ResNet50`, `EfficientNet-B0`, `DenseNet121`, `MobileNetV3-Large`, `ViT-B/16`.
- The four listed first gave the best results.
- Each image is CLAHE-normalized, replicated into 3 channels, resized to 224×224 and standardized with ImageNet mean/std.

Analytical objectives

Our methods address several analytical tasks on the tear-desiccate images:

- **Pattern recognition**: identifying class-specific morphological patterns (branching dendrites, dot clusters, cypress-like structures).
- **Recurring structure identification**: KMeans segmentation exposes recurring region types (background, figure, bright dots).
- **Similarity detection**: duplicate / near-duplicate detection during dataset cleaning.
- **Data comparison**: inter-class comparison of gradient statistics (HOG) and region statistics (KMeans).
- **Feature learning**: CNN backbones learn high-level representations that encode correlations between fine-grained texture patterns and the underlying chronic condition.

Task and expected outcome

- **Goal**: a classifier that, given a tear-desiccate image, estimates the probability that a patient has a specific **chronic disease**.
- **Evaluation**: a hidden test set; ranking by **weighted F1-score**.
- **Winner**: the team with the highest weighted F1 wins the track.

Results

#	Model	CV F1 (weighted)	Test F1 (weighted)
1	**ResNet50** *(winner)*	**0.8220**	**0.7015**
2	ResNet18	0.7973	0.5854
3	DenseNet121	0.7217	0.5834
4	EfficientNet-B0	0.7096	0.6849
5	KMeans (K=3) + SVM	0.6974	0.5582
6	HOG + SVM	0.4916	0.5155

Deep transfer learning clearly outperforms classical pipelines. **ResNet50** is our final submitted model.

Conclusion

- Careful preprocessing (ROI crop, grayscale, CLAHE, rotation augmentation, outlier removal) was crucial given the small dataset.
- Classical HOG/KMeans pipelines provide interpretable baselines but are limited by the handcrafted features.
- Transfer learning with pretrained CNNs dominates both CV and test performance; **ResNet50** achieves the best weighted F1.
- Future work: more data, ensembling across backbones, self-supervised pretraining on raw tear images.

Thank you!

Questions?

Luka Divjaković & Veselin Roganović
HackKošice – Pavol Jozef Šafárik University in Košice track